

Executive Brief: Tasty Treats Case

The Total Acquisition Cost (TAC) for Tasty Treats was done with a focus on the initial values mentioned by Swink et al (2020). The existing TAC is based on an order quantity of 1,000 units at a product cost of \$25. The formula for TAC is:

TAC=Annual Order Cost+Annual Carrying Cost+Annual Product

1. Existing TAC Calculation:

- **Annual Demand (D):** 12,000 units
- **Order Cost (S):** \$150
- **Carrying Cost Rate (H):** 20% of product cost
- **Product Cost (C):** \$25 per unit
- **Annual Order Cost:**

$$\text{Annual Order Cost} = \left(\frac{D}{Q}\right) \times S = \left(\frac{12,000}{1,000}\right) \times 150 = 18,000$$

$$\text{Average Inventory} = \frac{Q}{2} = \frac{1,000}{2} = 500$$

$$\text{Annual Carrying Cost} = \text{Average Inventory} \times H \times C = 500 \times 0.20 \times 25 = 2,500$$

$$\text{Annual Product Cost} = D \times C = 12,000 \times 25 = 300,000$$

$$\text{Total TAC} = 18,000 + 2,500 + 300,000 = 304,300$$

2. TAC After EOQ Calculation (Product Cost \$25):

$$\text{Economic Order Quantity (EOQ)} = \sqrt{\frac{2DS}{H \times C}} = \sqrt{\frac{2 \times 12,000 \times 150}{0.25 \times 25}} = \sqrt{360,000} = 600$$

$$\text{New Annual Order Cost} = \left(\frac{D}{Q}\right) \times S = \left(\frac{12,000}{600}\right) \times 150 = 3,000$$

$$\text{New Average Inventory} = \frac{Q}{2} = \frac{600}{2} = 300$$

$$\text{New Annual Carrying Cost} = 300 \times 0.20 \times 25 = 1,500$$

$$\text{New Total TAC} = 3,000 + 1,500 + 300,000 = 304,500$$

3. TAC After EOQ Calculation (Product Cost \$24.50):

$$\text{New EOQ} = \sqrt{\frac{2DS}{H \times C}} = \sqrt{\frac{2 \times 12,000 \times 150}{0.20 \times 24.5}} = 210$$

$$\text{New Annual Order Cost} = \left(\frac{D}{Q}\right) \times S = \left(\frac{12,000}{210}\right) \times 150 = 2,938$$

$$\text{New Average Inventory} = \frac{Q}{2} = \frac{210}{2} = 105$$

$$\text{New Annual Carrying Cost} = 105 \times 0.20 \times 24.50 = 1,490$$

New Total TAC = $2,938 + 1,490 + 294,000 = 298,428$

4. Safety Stock Policy Cost:

Assuming the safety stock is maintained to achieve a 95% service level, we calculate safety stock as $SS = z \times \sigma L$ where z is the z-score for a 95% service level (1.645) and σL is the standard deviation of demand during lead time.

With a reduction in TAC due to a better EOQ and safety stock policy, I would recommend taking the discount, as the overall TAC would decrease significantly, providing better financial leverage. A service level of 95% should be offered to balance customer satisfaction with inventory costs. This will help reduce cases of stockouts and ensure timely delivery, hence boosting the supply chain efficiency.

In conclusion, according to the above-analyzed inventory strategies, it will be possible to increase the utilization of supply chain management, decrease carrying costs, and reach better customer demand satisfaction at Tasty Treats.